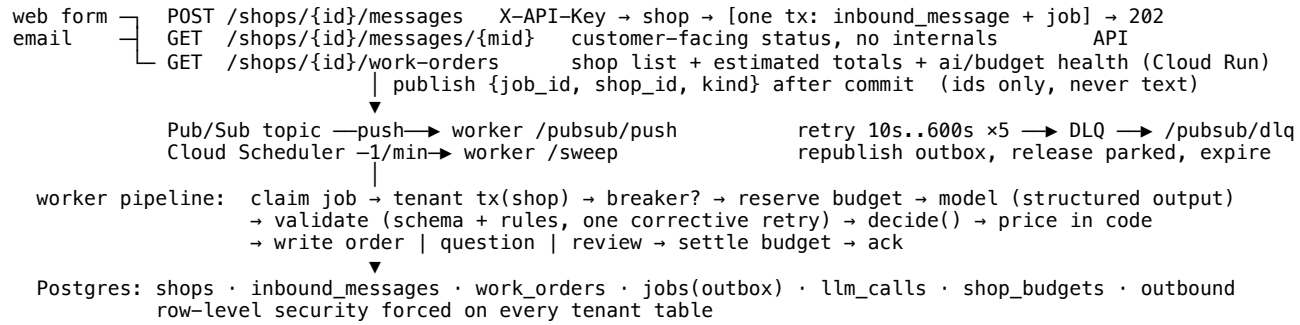
Architecture note

Riverbend work orders turns free-text customer messages into priced, structured work orders for three bike shops that must never see each other's data. Decision records with the alternatives considered are in `decisions.md`; this note is the shape, the boundaries, the data model, tenancy, and the three decisions I most want challenged.

Shape



Two processes from one image. The API never calls the model. The worker accepts input only from Pub/Sub and Scheduler.

Module boundaries

Module	Owns	Must not
`api/`	HTTP contract; tenant from the API key	touch the model; trust the body for identity
`ingest.py`	the accept transaction (message + outbox job); publish after commit	block on the broker
`dispatch.py`	moving a job id to the worker (Pub/Sub, inline, doubles)	carry customer text
`worker/pipeline.py`	the stages above; retry and park policy	compute a price or a date
`extraction/`	prompt, model schema, validation rules, provider interface (Azure, fake)	decide policy
`decide.py`, `pricing.py`, `dates.py`	pure: ask-or-order policy, arithmetic, date resolution	call anything
`repo/`	SQL per table; tenant tables reachable only inside `tenant_tx`	cross-table policy

The line that matters most: **the model proposes, code disposes.** The model returns catalog codes, quantities, a classified date phrase, and a proposed outcome. Everything that costs money or changes state is computed or decided by tested code.

Data model

`shops` (id, api_key_hash, timezone, monthly_cap_micros) is the tenant; every other table starts with `shop_id`. `inbound_messages` is unique on `(shop_id, message_id)`, the idempotency key the sender controls. `work_orders` is unique on `(shop_id, source_message_id)`, so a job that runs twice is a no-op. Foreign keys are composite, so an order in one shop cannot reference a message in another even by accident. `jobs` is the outbox and the processing record (status, attempts, run_after, park_reason) and holds ids only. `llm_calls` is the ledger: tokens, cost from a config price table, latency, outcome, validation errors, raw output. `shop_budgets` holds spent and reserved per month.

A work order moves `processing → open | awaiting_customer | needs_review`, then `awaiting_customer → (reply) → open | needs_review`, or `→ expired` after seven days.

Output that fails validation never becomes a row; the failure is recorded on the message.

Tenancy

Three layers, each catching a different mistake. (1) The API key resolves the shop and must match the path; a valid key on another shop's path is a 404, so shops cannot be enumerated. (2) Composite keys make cross-shop references unrepresentable. (3) Row-level security is enabled and *forced* on every tenant table, keyed on a transaction-local setting. A handler that forgets its WHERE clause gets only its own rows; a path that forgets to set the tenant gets zero rows and can write none. The worker claims the job across shops (ids only), then opens a tenant transaction for everything else, so even the prompt builder cannot read another shop's text. Tests prove each layer at the SQL and HTTP level.

Failure handling

Duplicates. The second arrival hits the unique constraint and returns the existing status with a 200; concurrent duplicates are resolved by the database. A redelivered job finds the order exists and acks without a second model call.

Provider slow or down. SDK retries are off. A timeout or 5xx returns 503 and Pub/Sub redelivers with backoff, five times, then dead-letters into a failed outcome. Three consecutive failures open a breaker; while open, jobs are parked with a run-after instead of burning attempts, and the sweeper republishes them. The customer sees "received, confirming shortly" throughout; the shop sees `ai.degraded`, the number waiting, and the number parked.

Spend cap. One conditional UPDATE reserves the worst case (two calls at max output) against the shop's month before each call and settles to actual after. At the cap the job parks until next month or until the cap is raised; the cap is read from `shops` on every reservation so a raise applies at once. The customer sees "received"; the shop sees `budget.exhausted`. Every call, including failed ones, is in the ledger, which is where the results table's spend comes from.

Three decisions to challenge

1. **Pub/Sub for delivery, Postgres outbox as the truth.** The assistant proposed a Postgres-only SKIP LOCKED queue; I overrode it (`ai-decisions-log.md`). I kept the outbox because the dual-write problem is real, and added the broker for push delivery, backlog scaling, and dead-lettering. The cost is a third system locally and job state in two places: Pub/Sub for normal retries, the row for parked states. Is that split clearer or murkier than one mechanism?
2. **No agent framework: a flat structured-output schema plus a pure `decide()`.** The one-question rule is an integer on the row, not a prompt instruction; history on resume is three stored rows replayed as chat turns. At what point does this stop being enough? My answer is when the conversation becomes open-ended and tool-driven. Someone may think that point is closer than I do.
3. **Row-level security in one shared schema rather than schema-per-shop.** Tenant scoping becomes a database guarantee rather than a code convention, at almost no cost at this size. But the worker and sweeper run under one role and switch tenants per transaction. A reviewer may prefer a role per shop or per-shop schemas for auditability at 100 shops.